# IGCSE 0607 Practice Questions — E1.3

## Powers and roots

**Total Questions:** 6 — **Total Marks:** 34 Marks

---

## Table of Contents & Quick Navigation


- **E1.3** Powers and roots
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*


---

**1** **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

**(b)** Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

**2** **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

**(b)** Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

**(c)** Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

**3** Solve the following equations.

**(a)**

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

**(b)**

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

**(c)**

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

**4** **(a)** Find the integer $k$ such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

**(b)** Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

**(c)** Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of $y^2$.

..... **[3]**

**5** **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

**(b)** Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

**(c)** Find the largest integer $n$ such that $n^4 < 2000$.

$n =$ ..... **[2]**

**6** **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

**(ii)** Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

**(b)** Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**